

Closed forms for the constants in the odd-weight identities between dihedral modular graph functions

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Abstract We study the combinations X_w of dihedral modular graph functions $C(a, b, c)$ of odd weight w introduced by D'Hoker, Green and Vanhove, whose Laplacian is proportional to the Eisenstein series $E(w)$. For the odd weights $3 \leq w \leq 61$ we show $X_w = f_w E(w) + g_w \zeta(w)$ with $f_w = 3((w-1)/2)!/w$ and $g_w = 6|B_{w-1}|/((w-1)/2)!$. The proof has three steps. The Laplace equation is derived in exact rational algebra. Harmonicity, invariance and polynomial growth force $X_w - f_w E(w)$ to be constant. The constant is read off from a Laurent polynomial evaluated in rational arithmetic. The space of combinations with $\Delta X \in \mathbb{Q}E(w)$ is shown exactly to be one-dimensional for odd w and zero for even w , for $3 \leq w \leq 25$. Independently of the proof, the verifier confirmed the constants numerically at $w = 3, \dots, 17$. The full weight-11 basis carries exactly one relation. The statement for all odd w is a conjecture. The proof rests on two assumptions not checked by code, a published theorem and a standard lemma.

Keywords Modular graph functions · Laplace eigenvalue equations · Eisenstein series · Bernoulli numbers · Genus-one string amplitudes · Computer-assisted proof

1 Introduction

Dihedral modular graph functions $C(a, b, c)$ arise as coefficients in the low-energy expansion of the one-loop closed-string amplitude. They satisfy Laplace eigenvalue equations with inhomogeneous terms that are polynomial in non-holomorphic Eisenstein series [1]. At each odd weight there is a combination X_w with Laplace eigenvalue zero, up to a term proportional to $E(w)$. Its multiplicity is one for each odd weight and zero for each even weight [1].

D'Hoker, Green and Vanhove determine such identities up to additive constants of integration, which are fixed by the behaviour near the cusp [1]. The constants g_7, g_9 are left undetermined there, while g_7 appears explicitly in a later work [2]. The Laurent polynomial of each $C(a, b, c)$ is known in closed form from D'Hoker and Kaidi. The open question is the value of f_w and g_w at larger odd w .

This is hard because the number of functions $C(a, b, c)$ grows with w , and the constant is a Laurent term of a long combination. At $w = 17$ the identity involves eight functions, with a leading integer coefficient 51819264000.

Numerical identification of the constants by integer-relation search is therefore fragile.

We prove closed forms for f_w and g_w by an exact route, and confirm them by independent numerics.

Summary of results.

- Theorem 1 shows $X_w = f_w E(w) + g_w \zeta(w)$ with closed forms for f_w, g_w , for the odd weights $3 \leq w \leq 61$ (exact Laplace algebra plus a rational Laurent computation).
- Theorem 2 shows that the space of combinations X with $\Delta X \in \mathbb{Q}E(w)$ is one-dimensional for odd w and zero for even w , for $3 \leq w \leq 25$ (exact).
- Conjecture 1 extends the closed forms beyond the proved range $w \leq 61$; it was read off from $w = 3, \dots, 13$ before the proof.
- The supporting propositions and numerical observations record independent confirmation at $w = 3, \dots, 17$, the full weight-11 relation space, and a hardening run.

2 Setting

Let $\tau = \tau_1 + i\tau_2$ and $p = m\tau + n$. The expansion variable of holomorphic forms is $q = e^{2\pi i\tau}$. The lattice sums are normalised as in D'Hoker–Green–Vanhove. $E(s)$ is the sum over nonvanishing lattice vectors p of $(\tau_2/(\pi|p|^2))^s$. $C(a, b, c)$ is the sum over triples of nonvanishing lattice vectors p_1, p_2, p_3 that add up to zero, of the product $\prod_i (\tau_2/(\pi|p_i|^2))^{a_i}$ with $(a_1, a_2, a_3) = (a, b, c)$. The weight is $w = a + b + c$. We write $E_w = E(w)$. ζ is the Riemann zeta function and B_n are Bernoulli numbers. The hyperbolic Laplacian is $\Delta = \tau_2^2(\partial_{\tau_1}^2 + \partial_{\tau_2}^2)$, acting as $\Delta E(s) = s(s-1)E(s)$. For odd w , X_w is the rational combination of $C(a, b, c)$ of weight w of D'Hoker–Green–Vanhove. The constants are

$$f_w = \frac{3((w-1)/2)!}{w}, \quad g_w = \frac{6|B_{w-1}|}{((w-1)/2)!}.$$

Numerical parameters of the verifier: working precision 32 digits; tolerance 1e-24 for relations (1e-14 with a Laplacian); fourth-order finite-difference Laplacian with step $h = 1e-6$; 4 random evaluation points (seed 4711) with $|\tau_1| \leq 1/2$ and $1 \leq \tau_2 \leq 2.5$. Exact criteria use the leading Laurent coefficient in $y = \pi\tau_2$.

Statements are labelled as proved (exact arithmetic), computer-assisted, or numerical (point evaluations).

Table 1 constants f_w and g_w in $X_w = f_w E(w) + g_w \zeta(w)$ (exact).

w	f_w	g_w
3	1	1

3 Main results

Theorem 1 (Closed forms, $w \leq 61$). *For the odd weights $3 \leq w \leq 61$, $X_w = f_w E(w) + g_w \zeta(w)$, with f_w and g_w as in the Setting.*

Proof. The argument has three steps, following.

1. Exact rational computation in the algebraic Laplace representation gives $\Delta X_w = w(w-1)f_w E(w)$ with no other terms. Since $\Delta E(w) = w(w-1)E(w)$, it follows that $\Delta(X_w - f_w E(w)) = 0$.
2. $X_w - f_w E(w)$ is harmonic, modular invariant and of polynomial growth at the cusp, hence constant.
3. The constant equals the τ_2^0 term of its Laurent polynomial. This polynomial was computed in rational arithmetic from Proposition 2.1 and Theorem 5.1 of D'Hoker–Kaidi, with the transcription checked against their eq. 5.19. The terms not involving products of zeta values agree with those of $f_w E(w) + g_w \zeta(w)$, which serves as a consistency check, and the constant term equals $g_w \zeta(w)$.

The verifier computed exactly at the odd weights $3 \leq w \leq 61$. Two assumptions are not checked by code: the correctness of Theorem 5.1 of D'Hoker–Kaidi (proved there; its conjectural part concerns only the coefficient of τ_2^{2-w} and is not used) and the standard lemma in step 2. $\square$

Novelty status: partly known. The combinations X_w for odd w are given in D'Hoker–Green–Vanhove (eq. 3.57), f_w for $w \leq 9$ and g_w for $w = 3$ and $w = 5$ in their eq. 3.34, with g_7, g_9 left undetermined. g_7 also appears explicitly in a later work [2]. Closed forms of f_w and g_w at larger odd w and their proof for $w \geq 9$ are not given there.

Theorem 2 (Uniqueness of the harmonic combination, $w \leq 25$). *For the weights $3 \leq w \leq 25$, the rational combinations X of dihedral $C(a, b, c)$ of weight w with $\Delta X \in \mathbb{Q}E(w)$ form a space of dimension 1 for odd w and 0 for even w . For odd w it is spanned by the combination of D'Hoker–Green–Vanhove (eq. 3.57), and $\Delta X = w(w-1)f_w E(w)$ in that normalisation.*

Proof. The space was determined exactly in the algebraic Laplace representation, in rational arithmetic, for the weights $3 \leq w \leq 25$. $\square$

Novelty status: partly known. The dimensions are stated in D'Hoker–Green–Vanhove. The closed form $f_w = 3((w-1)/2)!/w$ is not given there, and our exact confirmation for $3 \leq w \leq 25$ is an independent recomputation.

∴ conjecture [Odd weights] For odd $w \geq 3$, $X_w = f_w E(w) + g_w \zeta(w)$ with f_w, g_w as in the Setting.

Status. This is a conjecture. It was read off from $w = 3, \dots, 13$ before the proof. It is proved for the odd weights $w \leq 61$ (Theorem 1). Beyond $w = 61$ it is not covered by the proof.

4 Supporting computations

Proposition 1 (Laplace equations at $w = 9$ and $w = 11$). *Exactly, $\Delta(9C(4, 4, 1) + 18C(4, 3, 2) + 4C(3, 3, 3)) = 288 E(9)$ and $\Delta(2C(5, 5, 1) + 4C(5, 4, 2) + 2C(5, 3, 3) + 3C(4, 4, 3)) = 100 E(11)$.*

Certificate. The algebraic Laplace representation of D'Hoker–Green–Vanhove was re-derived and checked in rational arithmetic; the verifier returned $\lambda = 288$ and $\lambda = 100$ respectively.

Novelty status: already known for $w \leq 9$ (D'Hoker–Green–Vanhove, eq. 3.33). The $w = 11$ combination is the formula of eq. 3.57; f_w, g_w are not given there.

Numerical observation 1 (Relations at $w = 3, 5, 7$). Numerically, as functions of τ : $C(1, 1, 1) - E(3) - \zeta(3) = 0$; $30C(2, 2, 1) - 12E(5) - \zeta(5) = 0$; $252C(3, 3, 1) + 252C(3, 2, 2) - 108E(7) - \zeta(7) = 0$.

Certificate. The exact leading Laurent coefficient vanishes, and the maximal relative residual at 4 verifier points (seed 4711, 32 digits) is 5.13e-44, 1.57e-43 and 1.59e-43 respectively, against the bound 1e-24.

Novelty status: already known for $w = 3$ and $w = 5$ [1] and for $w = 7$ [2].

Numerical observation 2 (Relation at $w = 9$). Numerically, $960E(9) + \zeta(9) - 2160C(4, 4, 1) - 4320C(4, 3, 2) - 960C(3, 3, 3) = 0$ as functions of τ .

Certificate. The exact leading Laurent coefficient in y^9 vanishes; the maximal relative residual at 4 verifier points is 8.33e-44, against the bound 1e-24.

Novelty status: partly known. The identity with an undetermined constant g_9 is in D'Hoker–Green–Vanhove (eq. 3.34); the value of g_9 is not given there.

Numerical observation 3 (Relations at $w = 11, 13, 15, 17$). Numerically, as functions of τ , the weight-11 combination $19008C(5, 5, 1) + 38016C(5, 4, 2) + 19008C(5, 3, 3) + 28512C(4, 4, 3) - 8640E(11) - \zeta(11)$ vanishes. The weight-13 identity has zeta term $-691\zeta(13)$. The weight-15 identity has zeta term $-\zeta(15)$. The weight-17 identity has zeta term $-3617\zeta(17)$.

Certificate. Each relation passed the exact leading-coefficient test and the point test at 4 verifier points. The maximal relative residuals are 1.21e-43 ($w = 11$), 5.67e-44 ($w = 13$), 6.46e-44 ($w = 15$) and 5.49e-44 ($w = 17$), against the bound 1e-24. The relation at $w = 15$ was a prediction made before its value was read, and the one at $w = 17$ a preregistered blind test.

Novelty status: partly known: existence and combination of the identity for odd w (eq. 3.57); f_w, g_w for $w \geq 11$ are not given there.

Numerical observation 4 (Full weight-11 basis). Among the ten functions $C(a, b, c)$ of weight 11, $E(11)$ and $\zeta(11)$, the rational linear relations form a space of dimension exactly 1, spanned by the weight-11 identity above.

Certificate. Completeness was certified by criterion 2: at 20 verifier points (seed 4713, $\tau_2 \in [0.9, 3]$) one singular value is below $1e-24$ (the largest small one is $1.11e-43$), and the other 11 exceed $1e-16$, the smallest being $3.69e-9$. The originally preregistered criterion 1 (15 points, seed 4712, the other singular values must exceed $1e-8$) failed: the smallest large singular value was $4.149e-9$. We report this openly. Criterion 2 was preregistered afterwards on new data (20 points, seed 4713) with threshold $1e-16$, which lies far above the kernel value $1.11e-43$ and below the smallest large singular value $3.69e-9$, so that it separates kernel from non-kernel on this matrix; it was passed.

Novelty status: partly known, as for the previous observation.

Numerical observation 5 (Hardening of the weight-11 identity). An independent direct lattice summation in double precision agrees to at most $1e-12$ relative at 8 of 8 points.

Certificate. At $\tau = (0.500000, 0.866026)$ the residual is $2.15e-42$ at 30 digits, $7.01e-57$ at 45 digits and $1.3e-71$ at 60 digits, and the independent lattice sum differs by $4.372e-14$. The run exposed an evaluator bug: the Fourier sum for E_s stopped when a term vanished, which happens at $\tau_1 = 1/4$. Before the fix that point showed a precision-independent residual of $3.5e-10$. All points reported were computed with the corrected code, and no earlier test point was affected.

Remark 1 (Status of the computer-assisted parts). The proofs of Theorems 1 and 2 rely on exact rational arithmetic (Laplace algebra and Laurent polynomial) executed by the verifier. They are not re-derived by hand. The trusted base is the Python implementation, including its rational arithmetic and its transcription of Theorem 5.1 of D’Hoker–Kaidi checked against their eq. 5.19, together with the two assumptions named in. The numerical observations additionally rely on 32-digit floating arithmetic at verifier-chosen points. A third party re-runs every certificate with `python -m asd.recheck modular` from the repository named under Declarations.

5 Discussion

The identities express, for odd w , the combination X_w of two-loop graphs through a single Eisenstein series plus a single zeta value, with rational constants of closed form. The exact route avoids numerical integer-relation searches, whose coefficient height is the main obstruction at large w (cf.). The numerical observations are independent confirmation, not part of the proof.

Limitations. Theorem 1 is conditional on the correctness of Theorem 5.1 of D’Hoker–Kaidi and on the standard lemma used in step 2. Both are cited, not checked

by code. The proved range is $w \leq 61$. The uniqueness statement of Theorem 2 is checked for $w \leq 25$.

What did not work. (a) A search for relations at weight 6 among $C(4, 1, 1)$, $C(3, 2, 1)$, $C(2, 2, 2)$ and products of Eisenstein series and zeta values produced no verified claim. (b) An exact table of levels for which eta quotients span $M_4(\Gamma_0(N))$ was not obtained in the logged attempts; it was obtained separately (Appendix A). (c) A direct integer-relation search over the weight-11 basis did not find the relation and was aborted; the likely reason is too low a precision relative to the coefficient height. The relation was later certified by singular values (criterion 2). (d) The attempts at the weight-13 and weight-15 constant formula produced no verified claim; the relations were verified directly later. (e) The first preregistered completeness criterion at weight 11 failed.

Open questions.

- (i) A proof for all odd w , presumably reducing to an identity between binomial sums.
- (ii) The constants in the non-harmonic eigenvalue equations, i.e. the zeta terms in the equations for $C(a, b, c)$ with eigenvalue $s(s-1)$, $s \neq 1$.
- (iii) The analogous statements at higher loop order.

Declarations

Affiliation Not stated in the author line supplied with this manuscript.

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Code and data availability Repository <https://github.com/NinjaTurtlesHackathons/Hacknation>, branch `claude/modulformen-paper`, directory `projects/modular`. Reproduce all certificates with `python -m asd.recheck modular` and rebuild the paper with `python -m asd.paper --domain modular`.

Competing interests None.

A Numerical methods and error control

Certificate types. An exact certificate computes the Laplace algebra and Laurent polynomials in rational arithmetic. A relation certificate (`mgf_relation`) first tests the exact leading Laurent coefficient in $y = \pi\tau_2$ and then the relative residual at verifier points (32 digits, seed 4711, tolerance $1e-24$, or $1e-14$ with a finite-difference Laplacian of step $1e-6$). Completeness of a relation space is certified by a singular-value gap of the normalised value matrix (seed 4712): one singular value below $1e-24$ and the others above $1e-8$.

Relation spaces at $w = 3, 5, 7, 9$. Numerically each has dimension exactly 1, spanned by the identities of the observations above. For weight 9 the smallest large singular value of the 12×9 matrix is $8.6e-7$.

Laplace equations. The finite-difference checks give $L[C(3, 1, 1)] - 6C(3, 1, 1) - \frac{86}{5}E(5) + 4E(2)E(3) -$

$\frac{1}{10}\zeta(5) = 0$, $L[C(2, 1, 1)] - 2C(2, 1, 1) - 9E(4) + E(2)^2 = 0$, $L[C(2, 2, 1)] - 8E(5) = 0$ and $L[C(2, 2, 1)] - 20C(2, 2, 1) + \frac{2}{3}\zeta(5) = 0$, with maximal relative residuals $1.63\text{e-}25$, $2.39\text{e-}25$, $6.99\text{e-}26$ and $6.74\text{e-}26$ respectively.

Verifier self-test (holomorphic identities). Each identity was proved with a Sturm certificate: the terms lie in the same space of weakly holomorphic forms, and the coefficients up to the stated order vanish exactly. The following identities passed.

- $\theta(1)^4 = \eta(2)^{20}\eta(1)^{-8}\eta(4)^{-8}$ on $\Gamma_0(4)$.
- $\theta(1)^2 = \eta(2)^{10}\eta(1)^{-4}\eta(4)^{-4}$ on $\Gamma_0(4)$.
- $\text{thetaE8} = E_4$.
- $\text{thetaE8}^2 = \text{thetaD16}$, $\text{thetaE8}^2 = E_4^2$ and $\text{thetaE8}^2 = E_8$.
- $1728\Delta = E_4^3 - E_6^2$.
- $j\Delta = E_4^3$.

Falsified variants (the factors 1727 and 1729 in place of 1728, and the q^1 -coefficient 481 of thetaD16 in place of 480) were rejected. Eta quotients with trivial character span $M_4(\Gamma_0(N))$ exactly for N in $[2, 4, 5, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 22, 24, 25, 26, 27, 28, 30]$ and not for N in $[1, 3, 7, 11, 13, 17, 19, 21, 23, 29]$ (exact, exhaustive for $N \leq 30$). The novelty status of the last result is “not found in a targeted search of 33 abstracts on 2026-10-04”.

Exact family checks. A weaker earlier run of the exact family check covered the odd weights examined, $3 \leq w \leq 35$. It is subsumed by Theorem 1.

Laboratory workflow. The laboratory ran 17 rounds with 37 verified statements and 4 negative results; every round was preregistered before the experiment. Attempts were made by agents named in the logs as `sparsam`, `numeriker`, `skeptiker` and `theoretiker`, plus a cascade run. Each claim was passed to an independent verifier. A red-team phase produced 53 adversarial counter-checks in total: 4 passed (none contradicting the statement), 48 did not pass (the altered claims were refuted), and 1 was not executable. Examples of refuted alterations are a doubled $\zeta(17)$ coefficient and $g_5 = 1/5$ in place of $1/10$.

B Provenance of the statements

The provenance table mapping every statement to its evidence follows (it is generated automatically).

The complete automatic checking protocol of the writing process is stored as `pruefprotokoll.json` in the project directory of the repository.

References

- [1] Eric D’Hoker, Michael B. Green, and Pierre Vanhove. On the modular structure of the genus-one Type II superstring low energy expansion. *arXiv:1502.06698*, 2015.
- [2] Daniele Dorigoni, Axel Kleinschmidt, and Oliver Schlotterer. Poincaré series for modular graph forms at depth two. I. Seeds and Laplace systems. *arXiv:2109.05017*, 2021.

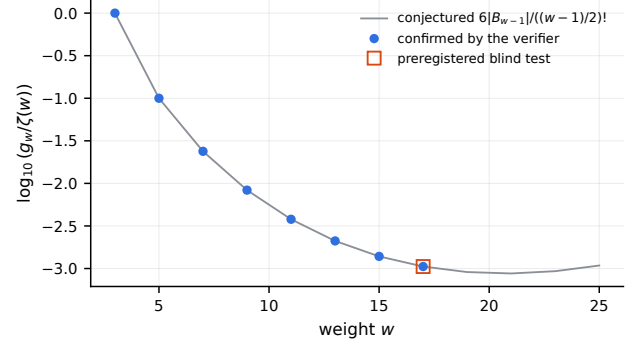


Fig. 1 Integration constants g_w of the odd-weight identities in the normalisation of eq. (3.57) of D’Hoker-Green-Vanhove: conjectured closed form (line) and the weights at which the verifier confirmed the identity numerically (dots, 32 digits, 4 verifier points); the square marks the preregistered blind test at $w = 17$.

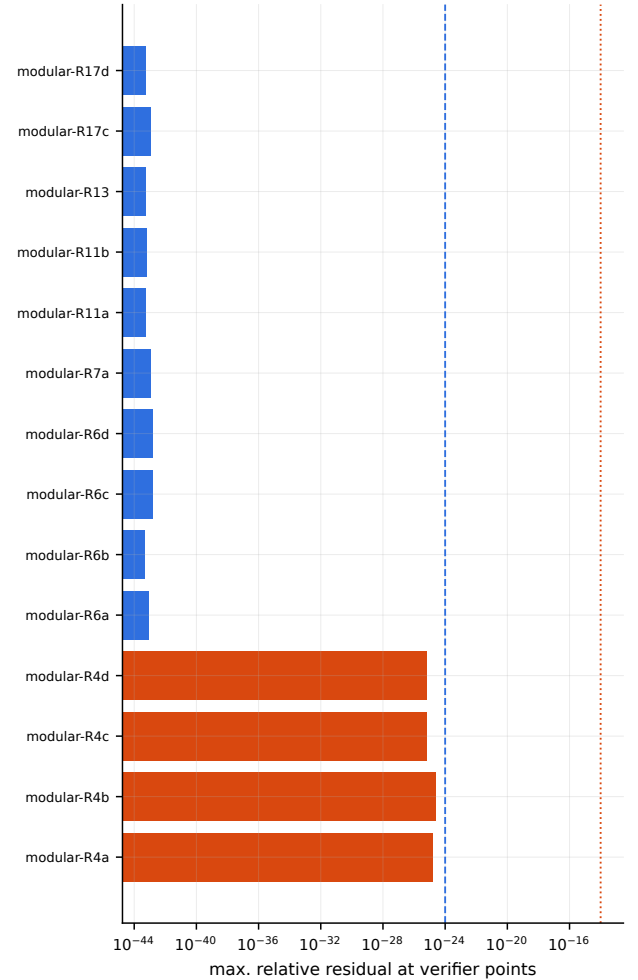


Fig. 2 Numerical verification margins: largest relative residual of each confirmed relation among modular graph functions at the verifier’s own random points (32 digits). Dashed: acceptance threshold 10^{-24} ; dotted: threshold 10^{-14} for relations containing a Laplacian (orange).

Statement	Level
Section: Introduction	computed_rigorous, observed
Section: Setting	computed_rigorous, observed
Theorem 1 (Closed forms, $w \leq 61$)	computed_rigorous
Section: Main results	computed_rigorous
Theorem 2 (Uniqueness of the harmonic combination, $w \leq 25$)	computed_rigorous
Proposition 1 (Laplace equations at $w = 9$ and $w = 11$)	computed_rigorous
Section: Supporting computations	computed_rigorous, observed
Observation 1 (Relations at $w = 3, 5, 7$)	observed
Observation 2 (Relation at $w = 9$)	observed
Observation 3 (Relations at $w = 11, 13, 15, 17$)	observed
Observation 4 (Full weight-11 basis)	observed
Observation 5 (Hardening of the weight-11 identity)	observed
Remark 1 (Status of the computer-assisted parts)	computed_rigorous, observed
Section: Discussion	computed_rigorous, observed
Section: Declarations	observed
Section: Appendix A: Numerical methods and error control	computed_rigorous, observed

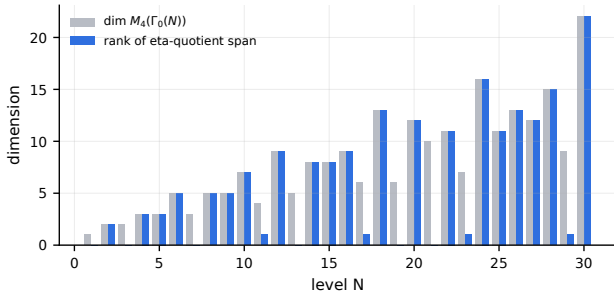


Fig. 3 Exact comparison, for every level $N \leq 30$, of $\dim M_4(\Gamma_0(N))$ (grey) with the rank of the span of all holomorphic eta quotients of weight 4 with trivial character (blue).